Inverse & Composite Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Composing functions**

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$(f \circ g)(x) = f(g(x)) = 2(x^2 - 1) + 3 = 2x^2 + 1$.
$(g \circ f)(x) = g(f(x)) = (2x + 3)^2 - 1 = 4x^2 + 12x + 9 - 1 = 4x^2 + 12x + 8$.

2 **[Foundational]** • **Target: Composing functions**

Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.

$(f \circ g)(2) = f(g(2)) = f(0) = \sqrt{0 + 4} = 2$. In general, $f \circ g$ requires $g(x) + 4 \geq 0$, i.e. $x^2 \geq 0$, which holds for all real x — so $f \circ g$ is defined everywhere.

3 **[Foundational]** • **Target: Decomposing a composite function**

Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g.

Let $g(x) = 3x - 5$ (inner function) and $f(x) = x^4$ (outer function). Then $f(g(x)) = (3x - 5)^4 = h(x)$.

4 **[Foundational]** • **Target: Decomposing a composite function**

Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

Let $g(x) = x^2 + 1$ and $f(x) = \sqrt{x}$. Then $f(g(x)) = \sqrt{x^2 + 1} = h(x)$.

5 **[Foundational]** • **Target: Verifying inverse functions**

Verify algebraically that $f(x) = \frac{x - 2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

$f(g(x)) = \frac{(3x + 2) - 2}{3} = \frac{3x}{3} = x$. $g(f(x)) = 3\left(\frac{x - 2}{3}\right) + 2 = (x - 2) + 2 = x$. Since both compositions return x, f and g are inverses.

6 **[Foundational]** • **Target: Verifying inverse functions**

Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.

Solve $y = 2x^3 + 1$ for x: $x^3 = \frac{y - 1}{2}$, so $f^{-1}(x) = \sqrt[3]{\frac{x - 1}{2}}$. Check at $x = 17$: $f^{-1}(17) = \sqrt[3]{8} = 2$, and $f(2) = 2(8) + 1 = 17$ ✓.

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7 [Foundational] • Target: Visualizing inverse functions as reflections

The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.

a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.

$f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.

b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically. Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

(a) $f(x) = x^2$ must be restricted to $x \geq 0$ (or $x \leq 0$) because the full parabola fails the horizontal line test — without restricting the domain, f is not one-to-one and has no inverse function.; (b) Setting $f(x) = x$: $x^2 = x$, so $x^2 - x = 0$, $x(x - 1) = 0$, giving $x = 0$ or $x = 1$. Both points $(0, 0)$ and $(1, 1)$ lie on $y = x$, matching where the two curves cross in the graph.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range of composite and inverse functions

Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = f(x + 1) = \frac{1}{(x+1)-3} = \frac{1}{x-2}$. This is undefined at $x = 2$, so the domain is all reals except $x = 2$.

9 [Intermediate] • Target: Domain and range of composite and inverse functions

If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1}.

The domain and range swap for an inverse function: f^{-1} has domain $[0, \infty)$ and range $[1, \infty)$.

10 [Intermediate] • Target: Applying composite and inverse functions in context

A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.

Solve $C = \frac{5}{9}(F - 32)$ for F: $\frac{9C}{5} = F - 32$, so $F = \frac{9}{5}C + 32$. This is C^{-1}, which converts a Celsius temperature back to Fahrenheit.

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11 [Intermediate] • Target: Using the horizontal line test

Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.

Since $f(x) = x^3 - 4x$ has multiple turning points (a cubic with two real critical points), a horizontal line such as $y = 0$ intersects the graph at more than one point ($x = 0, \pm 2$), failing the horizontal line test. So f does not have an inverse function over its entire domain.

12 [Intermediate] • Target: Using the horizontal line test

Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x, and find its inverse.

$f(x) = 5x - 2$ is a line with nonzero slope, so it is strictly increasing and every horizontal line crosses it exactly once — it passes the horizontal line test everywhere. Solving $y = 5x - 2$ for x: $f^{-1}(x) = \frac{x+2}{5}$.

13 [Intermediate] • Target: Composite functions with restricted domains

Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

$(f \circ g)(x) = f(x - 4) = \frac{1}{x-4}$. This is undefined at $x = 4$, so the domain is all reals except $x = 4$.

14 [Intermediate] • Target: Decomposing a composite function

Given $f(x) = 7x + 7$, find $f^{-1}(-21)$.

Final Answer: -4. Solve $-21 = 7x + 7$ for x: $x = \frac{-21-7}{7} = -4$, so $f^{-1}(-21) = -4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying inverse functions • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Given $f(x) = -5x + 8$, find $f^{-1}(23)$.

Final Answer: -3. Solve $23 = -5x + 8$ for x: $x = \frac{23-8}{-5} = -3$, so $f^{-1}(23) = -3$.

16 [Challenging] • Target: Visualizing inverse functions as reflections • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Given $f(x) = 2x - 9$, find $f^{-1}(-19)$.

Final Answer: -5. Solve $-19 = 2x - 9$ for x: $x = \frac{-19+9}{2} = -5$, so $f^{-1}(-19) = -5$.

17 [Challenging] • Target: Domain and range of composite and inverse functions • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Given $f(x) = 8x - 3$, find $f^{-1}(-3)$.

Final Answer: 0. Solve $-3 = 8x - 3$ for x: $x = \frac{-3+3}{8} = 0$, so $f^{-1}(-3) = 0$.

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18 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Given $f(x) = -9x + 4$, find $f^{-1}(4)$.

Final Answer: 0. Solve $4 = -9x + 4$ for x: $x = \frac{4-4}{-9} = 0$, so $f^{-1}(4) = 0$.

19 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Given $f(x) = 5x - 7$ and $g(x) = -x^2 + 5$, find $(g \circ f)(-3)$.

Final Answer: -479. $(g \circ f)(-3) = g(f(-3)) = g(-22) = -1(-22)^2 + 5 = -479$.

20 [Challenging] • Target: Composite functions with restricted domains • Cross-Lesson: 4.03 Parametric Equations & Modeling

Given $f(x) = -5x - 8$ and $g(x) = 5x^2 - 6$, find $(g \circ f)(-1)$.

Final Answer: 39. $(g \circ f)(-1) = g(f(-1)) = g(-3) = 5(-3)^2 - 6 = 39$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composing functions

Given $f(x) = x - 1$ and $g(x) = 5x - 4$, find $(f \circ g)(-6)$.

a. -34 **b.** -36

c. -35 ✓ **d.** -39

$(f \circ g)(-6) = f(g(-6)) = f(-34) = 1(-34) - 1 = -35$.

22 [Foundational] • Target: Decomposing a composite function

Given $f(x) = 5x - 2$ and $g(x) = x + 2$, find $(f \circ g)(-2)$.

a. -3 **b.** -10

c. -2 ✓ **d.** -1

$(f \circ g)(-2) = f(g(-2)) = f(0) = 5(0) - 2 = -2$.

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23 [Foundational] • Target: Verifying inverse functions

Given $f(x) = 9x + 5$ and $g(x) = -8x - 9$, find $(f \circ g)(-2)$.

a. 69

b. 67

c. 68 ✓

d. 95

$(f \circ g)(-2) = f(g(-2)) = f(7) = 9(7) + 5 = 68$.

24 [Foundational] • Target: Visualizing inverse functions as reflections

Given $f(x) = -9x + 2$ and $g(x) = -8x - 1$, find $(f \circ g)(6)$.

a. 442

b. 444

c. 415

d. 443 ✓

$(f \circ g)(6) = f(g(6)) = f(-49) = -9(-49) + 2 = 443$.

25 [Foundational] • Target: Domain and range of composite and inverse functions

Given $f(x) = 4x - 1$ and $g(x) = -7x - 3$, find $(f \circ g)(0)$.

a. -13 ✓

b. -14

c. 4

d. -12

$(f \circ g)(0) = f(g(0)) = f(-3) = 4(-3) - 1 = -13$.

26 [Foundational] • Target: Applying composite and inverse functions in context

Given $f(x) = 6x - 9$ and $g(x) = -x - 1$, find $(f \circ g)(5)$.

a. -46

b. -22

c. -45 ✓

d. -44

$(f \circ g)(5) = f(g(5)) = f(-6) = 6(-6) - 9 = -45$.

27 [Foundational] • Target: Using the horizontal line test

Given $f(x) = 6x + 4$ and $g(x) = 9x - 6$, find $(f \circ g)(1)$.

a. 84

b. 21

c. 23

d. 22 ✓

$(f \circ g)(1) = f(g(1)) = f(3) = 6(3) + 4 = 22$.

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28 [Foundational] • Target: Composite functions with restricted domains

Given $f(x) = -3x - 8$ and $g(x) = 4x - 5$, find $(f \circ g)(1)$.

a. -4

b. -6

c. -5 ✓

d. -49

$(f \circ g)(1) = f(g(1)) = f(-1) = -3(-1) - 8 = -5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Composition of Functions and Domain

Given $f(x) = 9x - 1$, find $f^{-1}(-73)$.

a. -8 ✓

b. -73

c. -7

d. -9

Solve $-73 = 9x - 1$ for x: $x = \frac{-73+1}{9} = -8$, so $f^{-1}(-73) = -8$.

30 [Intermediate] • Target: Finding Inverse Functions

Given $f(x) = -4x + 5$, find $f^{-1}(29)$.

a. -6 ✓

b. -7

c. -5

d. 29

Solve $29 = -4x + 5$ for x: $x = \frac{29-5}{-4} = -6$, so $f^{-1}(29) = -6$.

31 [Intermediate] • Target: Graphical Symmetry & Invertibility

Given $f(x) = 2x + 6$, find $f^{-1}(-4)$.

a. -6

b. -5 ✓

c. -4

d. -7

Solve $-4 = 2x + 6$ for x: $x = \frac{-4-6}{2} = -5$, so $f^{-1}(-4) = -5$.

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32 [Intermediate] • Target: Verifying Inverses Algebraically

Given $f(x) = -8x + 7$, find $f^{-1}(-25)$.

a. 4 ✓

b. 3

c. -25

d. 5

Solve $-25 = -8x + 7$ for x: $x = \frac{-25-7}{-8} = 4$, so $f^{-1}(-25) = 4$.

33 [Intermediate] • Target: Composing functions

Given $f(x) = 6x + 7$, find $f^{-1}(-11)$.

a. -11

b. -3 ✓

c. -4

d. -2

Solve $-11 = 6x + 7$ for x: $x = \frac{-11-7}{6} = -3$, so $f^{-1}(-11) = -3$.

34 [Intermediate] • Target: Decomposing a composite function

Given $f(x) = -2x - 1$ and $g(x) = 5x^2$, find $(g \circ f)(1)$.

a. 45 ✓

b. 43

c. -15

d. 47

$(g \circ f)(1) = g(f(1)) = g(-3) = 5(-3)^2 = 45$.

35 [Intermediate] • Target: Verifying inverse functions

Given $f(x) = -7x$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-7, 0)$

b. $\left(-\frac{1}{7}, 0\right)$ ✓

c. $\left(\frac{1}{7}, 0\right)$

d. $\left(\frac{5}{14}, 0\right)$

Solve $y = -7x$ for x: $x = \frac{y}{-7}$. So $f^{-1}(x)$ has slope $\frac{1}{-7} = -\frac{1}{7}$ and intercept 0: $\left(-\frac{1}{7}, 0\right)$.

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36 **[Intermediate]** • **Target: Visualizing inverse functions as reflections**

Given $f(x) = -9x - 1$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $\left(\frac{1}{9}, -\frac{1}{9}\right)$

b. $(-9, -1)$

c. $\left(-\frac{1}{9}, -\frac{1}{9}\right)$ ✓

d. $\left(-\frac{1}{9}, \frac{1}{9}\right)$

Solve $y = -9x - 1$ for x: $x = \frac{y+1}{-9}$. So $f^{-1}(x)$ has slope $\frac{1}{-9} = -\frac{1}{9}$ and intercept $-\frac{1}{9}$: $\left(-\frac{1}{9}, -\frac{1}{9}\right)$.

37 **[Intermediate]** • **Target: Domain and range of composite and inverse functions**

Given $f(x) = 6x - 5$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $\left(-\frac{1}{6}, \frac{5}{6}\right)$

b. $(6, -5)$

c. $\left(\frac{1}{6}, -\frac{5}{6}\right)$

d. $\left(\frac{1}{6}, \frac{5}{6}\right)$ ✓

Solve $y = 6x - 5$ for x: $x = \frac{y+5}{6}$. So $f^{-1}(x)$ has slope $\frac{1}{6} = \frac{1}{6}$ and intercept $\frac{5}{6}$: $\left(\frac{1}{6}, \frac{5}{6}\right)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Determine whether $f(x) = x^3 - 3x$ has an inverse function over its entire domain, using the horizontal line test.

a. Does NOT have an inverse function over its entire domain (not one-to-one) ✓

b. Has an inverse only on the interval $x \geq 0$

c. Cannot be determined without a graph

d. Has an inverse function over its entire domain (strictly increasing)

Since $b = -3 < 0$, $f(x) = x^3 - 3x$ has two turning points (critical points where $3x^2 - 3 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

39 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Determine whether $f(x) = x^3 - 2x$ has an inverse function over its entire domain, using the horizontal line test.

a. Does NOT have an inverse function over its entire domain (not one-to-one) ✓

b. Cannot be determined without a graph

c. Has an inverse function over its entire domain (strictly increasing)

d. Has an inverse only on the interval $x \geq 0$

Since $b = -2 < 0$, $f(x) = x^3 - 2x$ has two turning points (critical points where $3x^2 - 2 = 0$), so some horizontal lines cross the graph more than once. Does NOT have an inverse function over its entire domain (not one-to-one).

40 [Challenging] • Target: Composition of Functions and Domain • Cross-Lesson: 4.03 Parametric Equations & Modeling

Given $f(x) = 2x + 5$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$.

a. $2x^2 + 3$ ✓

b. $4x^2 + 20x + 24$

c. $2x^2 + 9$

d. $2x^3 - 1$

$(f \circ g)(x) = f(g(x)) = 2(x^2 - 1) + 5 = 2x^2 - 2 + 5 = 2x^2 + 3$.

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41 [Challenging] • Target: Composition of Functions and Domain • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

If $h(x) = (3x - 4)^5$, which two functions f and g satisfy $h(x) = (f \circ g)(x)$?

a. $f(x) = x^5$ **and** $g(x) = 3x - 4$ ✓

b. $f(x) = 3x - 4$ and $g(x) = x^5$

c. $f(x) = 3x^5$ and $g(x) = x - 4$

d. $f(x) = (x - 4)^5$ and $g(x) = 3x$

Let inner function $g(x) = 3x - 4$ and outer function $f(u) = u^5$. Then $f(g(x)) = (3x - 4)^5$.

42 [Challenging] • Target: Finding Inverse Functions • Cross-Lesson: 1.02 Rational Functions & Asymptotes

What is the inverse function of $g(x) = x^3 + 7$?

a. $g^{-1}(x) = \sqrt[3]{x - 7}$ ✓

b. $g^{-1}(x) = \sqrt[3]{x} - 7$

c. $g^{-1}(x) = (x - 7)^3$

d. $g^{-1}(x) = \frac{1}{x^3 + 7}$

$y = x^3 + 7 \Rightarrow y - 7 = x^3 \Rightarrow x = \sqrt[3]{y - 7}$. Swap variables: $g^{-1}(x) = \sqrt[3]{x - 7}$.

43 [Challenging] • Target: Graphical Symmetry & Invertibility • Cross-Lesson: 1.03 Rates of Change & Concavity

The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of each other across which line?

a. $y = x$ ✓

b. $y = -x$

c. $x = 0$ (y-axis)

d. $y = 0$ (x-axis)

Because the inverse interchanges the inputs and outputs $(a, b) \leftrightarrow (b, a)$, the graph of f^{-1} is the reflection of the graph of f across the line $y = x$.

44 [Challenging] • Target: Graphical Symmetry & Invertibility • Cross-Lesson: 2.01 Exponential Functions & Growth Models

If $f(2) = 7$ and $f'(2) = 4$, what is $(f^{-1})(7)$?

a. 2 ✓

b. 7

c. $\frac{1}{4}$

d. $\frac{1}{2}$

By definition of the inverse function, $f(a) = b \Leftrightarrow f^{-1}(b) = a$. Since $f(2) = 7$, $f^{-1}(7) = 2$.

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45 [Challenging] • Target: Verifying Inverses Algebraically • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Let $f(x) = 5x - 9$ and $g(x) = \frac{x+9}{5}$. Evaluate $f(g(x))$.

a. x ✓

b. $5x$

c. $x - 9$

d. 1

$f(g(x)) = 5\left(\frac{x+9}{5}\right) - 9 = (x + 9) - 9 = x$.

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